

PDF release of bom.xlsx

Sheet: BOM

Component	Mass (g/cell)	kg/kWh	Formula	Source
Positive active material	13.9296	0.7480114761413958	$d_{\text{pos}}(1-e_{\text{pos}})\rho_{\text{pos}}A = 75.6\mu\text{m} \cdot 0.55 \cdot 3262 \cdot 0.1027$	r5_v11_energy.json:layer_kg_m2 x area
Positive conductive additive (2 wt% of active, literature default)	0.2786	0.0149606591182082	$13.9296 \text{ g} \times 0.02$	literature default, annotated
Positive binder (2 wt% PVDF, literature default)	0.2786	0.0149606591182082	$13.9296 \text{ g} \times 0.02$	literature default, annotated
Negative active material	8.4093	0.4515745539222835	$d_{\text{neg}}(1-e_{\text{neg}})\rho_{\text{neg}}A = 85.2\mu\text{m} \cdot 0.58 \cdot 1657 \cdot 0.1027$	r5_v11_energy.json:layer_kg_m2 x area
Negative conductive additive (1 wt% of active, literature default)	0.0841	0.004516121435180578	$8.4093 \text{ g} \times 0.01$	literature default, annotated
Negative binder (2 wt% CMC/SBR, literature default)	0.1682	0.009032242870361156	$8.4093 \text{ g} \times 0.02$	literature default, annotated
Separator	0.1305	0.007007774640797449	$d_{\text{sep}}(1-e_{\text{sep}})\rho_{\text{sep}}A = 8\mu\text{m} \cdot 0.4 \cdot 397 \cdot 0.1027$	r5_v11_energy.json:layer_kg_m2.separator x area
Electrolyte (1.2 g/cm ³ literature density, fill factor 1.0)	9.1944	0.4937339705543913	pore volume x 1.2 = 7.662 cm ³ x 1.2	pore volume from geometry; density literature value
Positive current collector (Al 10 μm)	2.7729	0.1489031287468754	$10\mu\text{m} \times 2700 \times 0.1027$	r5_v11_energy.json:layer_kg_m2.positive_cc x area
Negative current collector (Cu 6 μm)	5.5212	0.2964852516993937	$6\mu\text{m} \times 8960 \times 0.1027$	r5_v11_energy.json:layer_kg_m2.negative_cc x area
Enclosure	Not modeled	Not modeled	—	honest omission
Tabs	Not modeled	Not modeled	—	honest omission
TOTAL (contract caliber, electrolyte excluded)	30.7634	1.651976815208492	sum of layer masses	r5_v11_energy.json:mass_kg
TOTAL incl. electrolyte + additives/binders	40.7666	2.189142878709067	sum of all rows above	mechanical sum
Cell energy (reference)	18.6222 Wh	—	$V \cdot I$ integral, 1C	r5_v11_energy.json:energy_wh